

Fares Elsobky

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Software Engineer at Airbus Defense & Space

Skills

Programming	Python Java JavaScript C++ SQL
Backend / Web	Spring Boot Spring Cloud Angular REST APIs OAuth 2.0
AI / Data	PyTorch TensorFlow Deep Learning Reinforcement Learning Data Analysis Bayesian Optimization
DevOps / Cloud	Docker Docker Compose Kubernetes GitLab CI/CD
Testing / Tools	Playwright REST Assured Git Jira
Other	SAP ABAP SAPUI5 OData Raspberry Pi PySceneDetect

Education

Technical University of Munich (TUM)

MASTER OF SCIENCE IN DATA ENGINEERING AND ANALYTICS

Munich, Germany

Graduation 2026

- Major: Data Engineering and Analytics | Minor: Software Engineering
- Master's Thesis: AI-Based Access Point Placement Optimization in Indoor Wireless Networks
- Developed a simulation-based optimization framework for indoor access point placement using Reinforcement Learning and Bayesian Optimization
- Built automated Python pipelines for experiments, benchmarking, and result analysis
- Evaluated performance using runtime, coverage, and feasibility metrics across multiple scenarios
- Improved optimization speed using TD3-based RL compared with traditional methods
- Applied machine learning, data analysis, and software engineering in a research environment

Data Engineering Analytics Software Engineering

Arab Academy for Science, Technology & Maritime Transport (AASTMT)

BACHELOR'S IN COMPUTING AND INFORMATION TECHNOLOGY

Cairo, Egypt

Graduation 2021

- Major: Computer Science (CS) - Software Engineering (SE)
- Bachelor Thesis (A+): Real-Time Car Violations Detection System
- Developed a real-time car violation detector using Raspberry Pi with cameras, IR, GPS, AI, OpenCV, and cloud computing
- Detected traffic-light, speed, and lane violations using OpenCV and automatically presented them on Azure
- Used the Google Maps API for location and speed, and built a Python Flask website connected to an Azure database for user access
- Reduced road-crash risks through advanced violation detection
- GPA: 3.4 (US GPA)

Computer Science Software Engineering Computer Vision

Professional Experience

Airbus Defense & Space

SOFTWARE ENGINEER: NOMINATION TOOL DEVELOPER

Munich, Germany

Oct 2022 - Oct 2024

- Developed and maintained an in-house Nomination Tool using Google Apps Script (JavaScript), achieving end-to-end digitization of nomination processes and significantly enhancing efficiency
- Tested each version before release to ensure reliability and performance, and handled user problems and errors to maintain a smooth user experience
- Deployed new releases and upgraded the tool from Version 6 to Version 11, continually adding features based on stakeholder requirements
- Integrated a new validation layer for nominations, automated data collection from Google Workspace, and streamlined the creation of forms and emails
- Improved the user experience through tool modularization and extensive automation, ensuring data coherence and supporting team operations
- Managed data coherence for the department, ensured accurate and timely data handling, and supported the team's daily operations
- Continually assessed and optimized tool performance, incorporating user feedback to improve efficiency and satisfaction

Google Apps Script JavaScript Automation

Projects

TUM & Capgemini

ADVANCED PRACTICAL COURSE - ENTERPRISE SOFTWARE ENGINEERING USING SAP

Munich, Germany

Oct 2023 - Feb 2024

- Implemented a sustainability cockpit using ABAP for backend functionality, OData for data exchange, and SAPUI5 for frontend visualization
- Met with stakeholders to collect project requirements and translate them into a product
- Created an Agile roadmap, data model, and software architecture
- Held weekly stakeholder meetings to present progress and refine the roadmap
- Built a company visualization dashboard with high-level reporting, recent-data tracing, and goal-achievement analysis

SAP ABAP OData SAPUI5

TUM - Data Processing

Munich, Germany

HIGH-LEVEL FEATURE EXTRACTION APPLICATION

Nov 2024 - May 2025

- Developed an application to extract high-level features from images using advanced machine-learning models
- Leveraged BLIP-2 for image captioning and integrated models from Hugging Face and Model Zoo, with plans for BLIP-3 video captioning and fine-tuning enhancements
- Implemented PySceneDetect to identify and select keyframes from video scenes
- Combined keyframes into composite images for comprehensive feature extraction
- Stored extracted high-level features and integrated the module with an existing tool

Machine Learning BLIP-2 PySceneDetect

TUM - Advanced Topics in Software Engineering

Munich, Germany

ONLINE IDE DEVELOPMENT

Oct 2024 - Apr 2025

- Collaborated on an online Integrated Development Environment as part of an advanced software engineering course
- Used Docker, Docker Compose, and GitLab CI/CD for virtualization, orchestration, and continuous integration
- Designed Angular component architecture and user interfaces, applying the Observer Pattern for distributed event handling
- Implemented Spring Cloud service discovery, API gateway patterns, load balancing, and a container-per-service approach
- Developed Spring Boot RESTful services, applied Spring Security, managed data access with JDBC/JPA/Hibernate, implemented OAuth 2.0, and conducted unit and integration testing

Angular Spring Boot Docker GitLab CI/CD

Languages _____

Arabic Native
English Fluent